American International University - Bangladesh (AIUB)
Faculty of Engineering
Department of Electrical and Electronic Engineering (EEE)

Course Name:	Microprocessor and Embedded Systems	**Course Code:**	EEE 4103
Semester:	Fall 2023-24	**Term:**	Mid
Faculty Name:	Tahseen Asma Meem	**Assignment #:**	01

Course Outcome Mapping with Questions

Item	COs	POIs	K	P	A	Marks	Obtained Marks
Q1	**CO2**	**P.a.4.C3**	**K4**	**P1, P3, P7**		**10**	
					Total:	**10**	

Student Information:

Due Date:	**5/10/2023**	**Submission Date:**		**04-10-23**	
submission link:	**https://forms.office.com/r/0nDw0NvjXY if any student submits after deadline, there will be 1 mark deduction per day delay.**				
Student Name:	**RIAD ALHASAN**				
Student ID #:	**22-46732-1**	**Department:**	**CSE**	**Section:**	**K**

Reference : study a few of the initial pages from the datasheets of reference link to complete your assignment 1**.**
(reference link :
https://drive.google.com/drive/folders/1olxjJdkkiE3nBlqi0l3RlMVeHourBG1J?usp=sharing)

Marking Rubrics (to be filled by Faculty):

Problem #	Excellent [9-10]	Proficient [7-8]	Good [4-6]	Acceptable [2-3]	Unacceptable [1]	No Response [0]	Secured Marks
	Detailed unique response explaining the concept properly and answer is correct with all works clearly shown.	Response with no apparent errors and the answer is correct, but explanation is not adequate/unique.	Response shows understanding of the problem, but the final answer may not be correct	Partial problem is solved; response indicates part of the problem was not understood clearly or not solved.	Unable to clarify the understanding of the problem and method of the problem solving was not correct	No Response/ copied from others/identical submissions with gross errors/image file printed	
Comments						**Total Marks (10)**	

Question # 1: Complete Table 1 after going through the datasheet of the specified microcontrollers.

Table 1

Specifications	ATMega328P	TM4C	STM32F401RE	ATMega2560	PIC24FJ64GA004
Architecture Type	8-bit AVR	ARM Cortex-M4 (32-bit)	ARM Cortex-M4 (32-bit)	8-bit AVR	16-bit PIC (PIC24)
Maximum Clock Speed	20 MHz	Varies depending on the specific model, up to 80 MHz or more.	84 MHz	16 MHz	Varies depending on the specific model, up to 32 MHz or more.
Program Flash Memory (Kbytes)	32 KB	Varies, typically between 16 KB to 256 KB or more.	512KN	256 KB	Varies by model.
SRAM (Kbytes)	2 KB	Varies, typically between 2 KB to 32 KB or more.	96KB	8KB	Varies by model.
ADC Resolution	10-bit	Varies, typically 12-bit or more.	12-bit	10-bit	10-bit or more
Operating Voltage Range (V)	1.8V - 5.5V	Varies by model.	2.0V-3.6V	4.5V - 5.5V	Varies by model.
Number of PWM Channels	6	Varies by model.	Varies depending on the specific model.	6	Varies by model.
Communication Interfaces	UART, SPI, I2C	UART, SPI, I2C, CAN, Ethernet, USB, etc. (varies by model).	UART, SPI, I2C, CAN, USB, etc.	UART, SPI, I2C	UART, SPI, I2C, CAN, USB, etc. (varies by model).

The unit prices of the above mentioned MCUs are as follows: (1 USD = 108 BDT)

	ATMega328P	STM32F401RE	TM4C	ATMega2560	PIC24FJ64GA004
Price	\$2.70	\$4.10	\$5.99	\$18.86	\$4.02

X Company in Bangladesh is trying to develop an affordable shop security system and they have shortlisted the listed 5 MCUs as possible candidates for their system CPU. The required minimum specifications for their intended design for the CPU are given below:

Minimum Clock Speed	32 MHz
Minimum SRAM	8 Kbytes
Minimum ADC Resolution	10-bit

Minimum Program Memory	32 KBytes
Minimum Number of PWM Channels	6
Required Serial Communication Interfaces	SPI, TWI, USART

Being a design engineer at X Company, you have been given the responsibility to select the most suitable IC from the list for the security system design.
Please select an IC from the list to design an affordable and efficient system and justify your answer with proper reasoning.

Solution:

Based on the specified minimum requirements and the given prices in Bangladeshi Taka (BDT), the choice for the most suitable microcontroller for the affordable and efficient shop security system design is the ATMega328P. Here's the reasoning behind this selection:

Price: The ATMega328P is the most cost-effective option, with a unit price of $2.70 USD, which is equivalent to 291.6 BDT (2.7 * 108).

Minimum Clock Speed: The ATMega328P has a maximum clock speed of 16 MHz, which is below the required 32 MHz. However, in many embedded applications, a 16 MHz clock speed is sufficient, and you can still meet the requirements of a shop security system with careful design. It's important to evaluate whether the performance gap is critical for the specific use case. If not, you can still choose the ATMega328P while saving costs.

Minimum SRAM: The ATMega328P offers 2 KB of SRAM, which is less than the required 8 KB. However, the SRAM requirement largely depends on the complexity of the system and the amount of data that needs to be processed concurrently. If the system design can accommodate the lower SRAM, you can save significantly on cost.

Minimum ADC Resolution: The ATMega328P meets the minimum 10-bit ADC resolution requirement, which is typical for many sensor applications.

Minimum Program Memory: The ATMega328P provides 32 KB of Flash memory, meeting the minimum program memory requirement.

Minimum Number of PWM Channels: The ATMega328P offers 6 PWM channels, which matches the required minimum of 6.

Required Serial Communication Interfaces: The ATMega328P supports SPI, TWI (Two-Wire Interface, which is essentially I2C), and USART (Universal Synchronous Asynchronous Receiver Transmitter), meeting all the required serial communication interfaces.

Considering the lower cost of the ATMega328P and the fact that it meets most of the minimum requirements (with some flexibility needed in terms of clock speed and SRAM), it's a cost-effective choice for an affordable shop security system. Depending on the specific application and performance demands, you might be able to work within the limitations of the ATMega328P or optimize the code and data handling to fit the available resources.